

Q1:



Q2:

Destination Port: 80

Q3:

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> Frame 373: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface en0, id 0
> Ethernet II, Src: aa:5d:13:55:5d:3a (aa:5d:13:55:5d:3a), Dst: Nokia_48:c9:4b (68:ab:09:48:c9:4b)
> Internet Protocol Version 4, Src: 100.64.18.176, Dst: 128.119.245.12
✓ Transmission Control Protocol, Src Port: 62194, Dst Port: 443, Seq: 0, Len: 0
  Source Port: 62194
  Destination Port: 443
  [Stream index: 9]
  [Stream Packet Number: 1]
  [Conversation completeness: Complete, WITH_DATA (63)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 4207198855
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1011 .... = Header Length: 44 bytes (11)
> Flags: 0x0c2 (SYN, ECE, CWR)
  Window: 65535
  [Calculated window size: 65535]
  Checksum: 0x632c [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (24 bytes) Maximum segment size, No Operation (NOP), Window scale, No Operation (NOP), No Operation (NOP)
0000  68 ab 09 48 c9 4b aa 5d 13 55 5d 3a 08 00 45 00  h..H.K.] .U]:..E.
0010  00 40 00 00 40 00 40 06 4e 44 64 40 12 b0 80 77  .@...@. NDd@...w
0020  f5 0c f2 f2 01 bb fa c4 c2 87 00 00 00 00 b0 c2  ....
0030  ff ff 63 2c 00 00 02 04 05 b4 01 03 03 06 01 01  ...c,.....
0040  08 0a 17 40 1d 61 00 00 00 00 04 02 00 00 00  ....@.a.....

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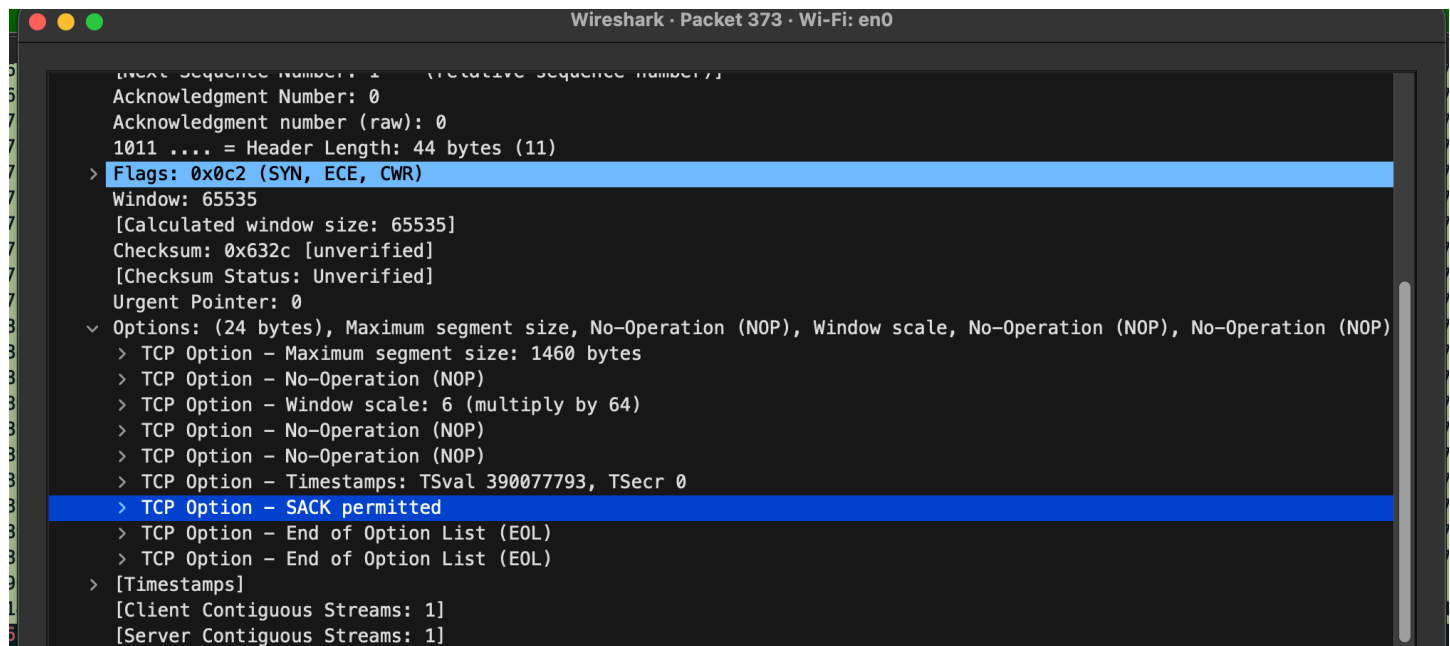
No.: 373 · Time: 8.401903 · Source: 100.64.18.176 · Destination: 128.119.245.12 · Prot...[SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=390077793 TSecr=0 SACK_PERM

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✓ Flags: 0x0c2 (SYN, ECE, CWR)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  .... 1... = Congestion Window Reduced: Set
  .... .1.. = ECN-Echo: Set
  .... ..0. = Urgent: Not set
  .... ...0 = Acknowledgment: Not set
  .... .... 0... = Push: Not set
  .... .... .0.. = Reset: Not set
> .... .... .1. = Syn: Set
  .... .... ...0 = Fin: Not set
  [TCP Flags: ....CE....S.]

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Sequence Number: 4207198855



SACK permitted

Q4:

Sequence Number: 4207198856

Q5:

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Internet Protocol Version 4, Src: 100.0.0.1, Dst: 10.0.0.1
  Transmission Control Protocol, Src Port: 62193, Dst Port: 80, Seq: 1, Ack: 1, Len: 591
    Source Port: 62193
    Destination Port: 80
    [Stream index: 7]
    [Stream Packet Number: 4]
    > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 591]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 3071775068
    [Next Sequence Number: 592 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 4258315001
    1000 .... = Header Length: 32 bytes (8)
    > Flags: 0x018 (PSH, ACK)
    Window: 2055
    [Calculated window size: 131520]
    [Window size scaling factor: 64]
    Checksum: 0xc0ec [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    > [Timestamps]
    > [SEQ/ACK analysis]
0020  f5 0c f2 f1 00 50 b7 17 91 5c fd d0 ba f9 80 18  ....P... \.....
0030  08 07 c0 ec 00 00 01 01 08 0a b3 da 86 09 b2 da  ....
0040  64 9b 50 4f 53 54 20 2f 77 69 72 65 73 68 61 72  d·POST / wireshar
0050  6b 2d 6c 61 62 73 2f 6c 61 62 33 2d 31 2d 72 65  k-labs/l ab3-1-re
0060  70 6c 79 2e 68 74 6d 20 48 54 54 50 2f 31 2e 31  ply.htm HTTP/1.1
0070  0d 0a 48 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e  ·Host: gaia.cs.
0080  75 6d 61 73 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65  umass.ed u·Conne
0090  63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76  ction: k eep-aliv
00a0  65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74  e·Conte nt-Lengt

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Sequence Number: 3071775068

The payload contains 591 bytes of data

No, the data did not fit into this single segment

Q6:

ACK Received Time for First Segment: 7.471139

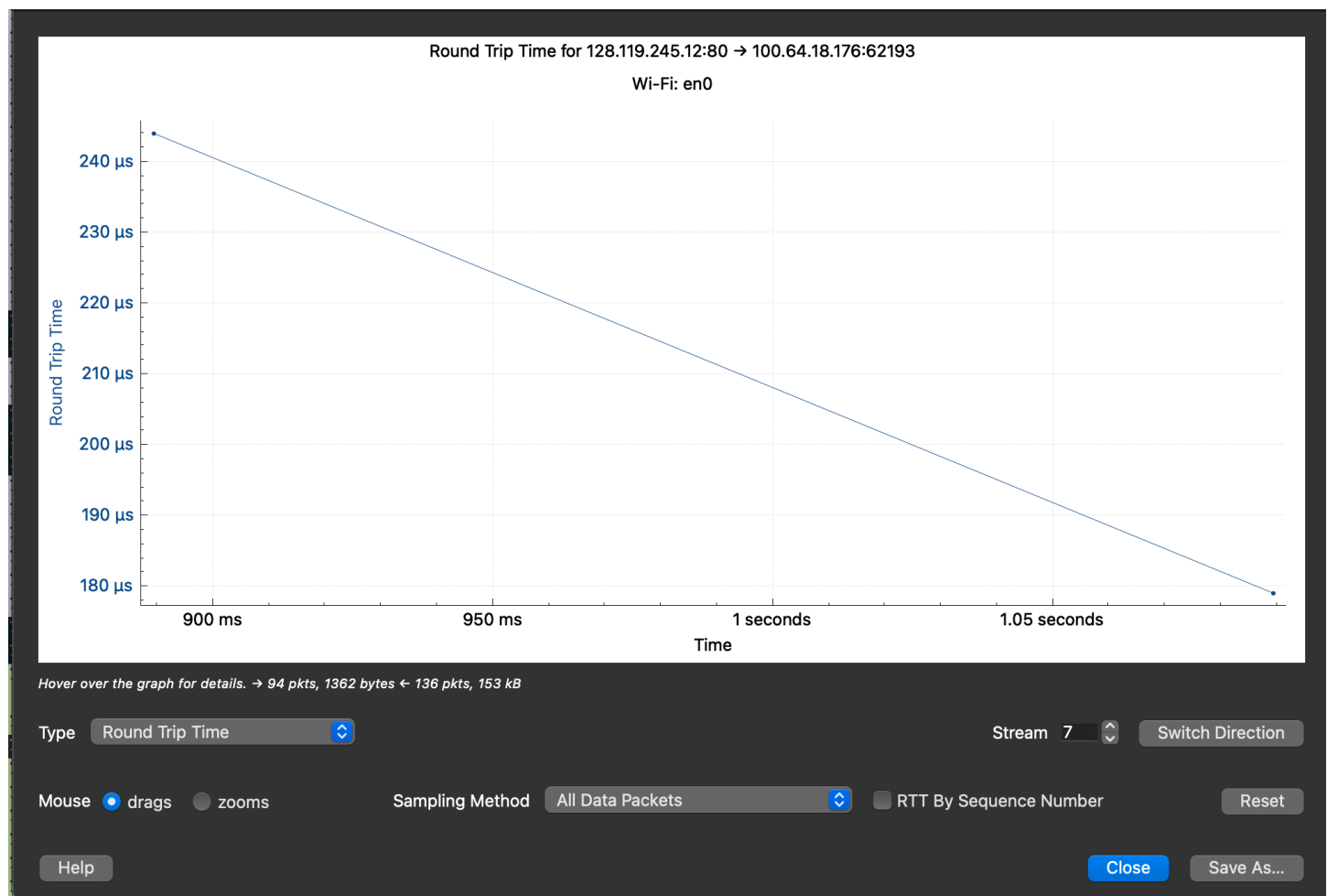
RTT for the First Data-Containing Segment: $RTT1 = 7.471139 - 7.448726 = 0.022413$

RTT for the Second Data-Carrying Segment: 0.022414

EstimatedRTT Calculation: using formula

$\alpha = 0.125$

EstimatedRTT = 0.022413125 seconds



Q7:

Segment 1 (Frame 138): 657 bytes (591 bytes payload + 32 bytes TCP header + 20 bytes IP header + 14 bytes Ethernet).

TCP segment size, 623 bytes (591 payload + 32 header).

Segment 2: 1514 bytes.

Segment 3: 1514 bytes.

Segment 4: 1514 bytes.

Q8:

Window: 2055

Calculated window size 131520

No the sender is not throttled as the buffer size is much larger than the data being sent

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> Frame 137: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface en0, id 0
> Ethernet II, Src: aa:5d:13:55:5d:3a (aa:5d:13:55:5d:3a), Dst: Nokia_48:c9:4b (68:ab:09:48:c9:4b)
> Internet Protocol Version 4, Src: 100.64.18.176, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 62193, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
  Source Port: 62193
  Destination Port: 80
  [Stream index: 7]
  [Stream Packet Number: 3]
> [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 3071775068
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 4258315001
  1000 .... = Header Length: 32 bytes (8)
> Flags: 0x010 (ACK)
Window: 2055
  [Calculated window size: 131520]
  [Window size scaling factor: 64]
  Checksum: 0x3c68 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0

0000  68 ab 09 48 c9 4b aa 5d 13 55 5d 3a 08 00 45 00  h..H.K.] .U]:.E.
0010  00 34 00 00 40 00 40 06 4e 50 64 40 12 b0 80 77  .4..@.@.NPd@...w
0020  f5 0c f2 f1 00 50 b7 17 91 5c fd d0 ba f9 80 10  ....P.. \.....
0030  08 07 3c 68 00 00 01 01 08 0a b3 da 86 08 b2 da  ..<h....
0040  64 9b                                     d.
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Q9:

Yes

312	7.99401b	100.64.18.176	128.119.245.12	HTTP	941	PUSH /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
313	7.994275	100.64.18.176	100.64.1.45	TCP	78	[TCP Retransmission] 7800 → 62919 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=64 TSval=1607962300 TSecr=3017443371
314	7.997714	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=73380 Win=141568 Len=0 TSval=3000657592 TSecr=3017443183

Q10:

Typical Amount Acknowledged: 1174

Yes, you can identify cases of Delayed ACK as there is some major differences between bytes of different segments

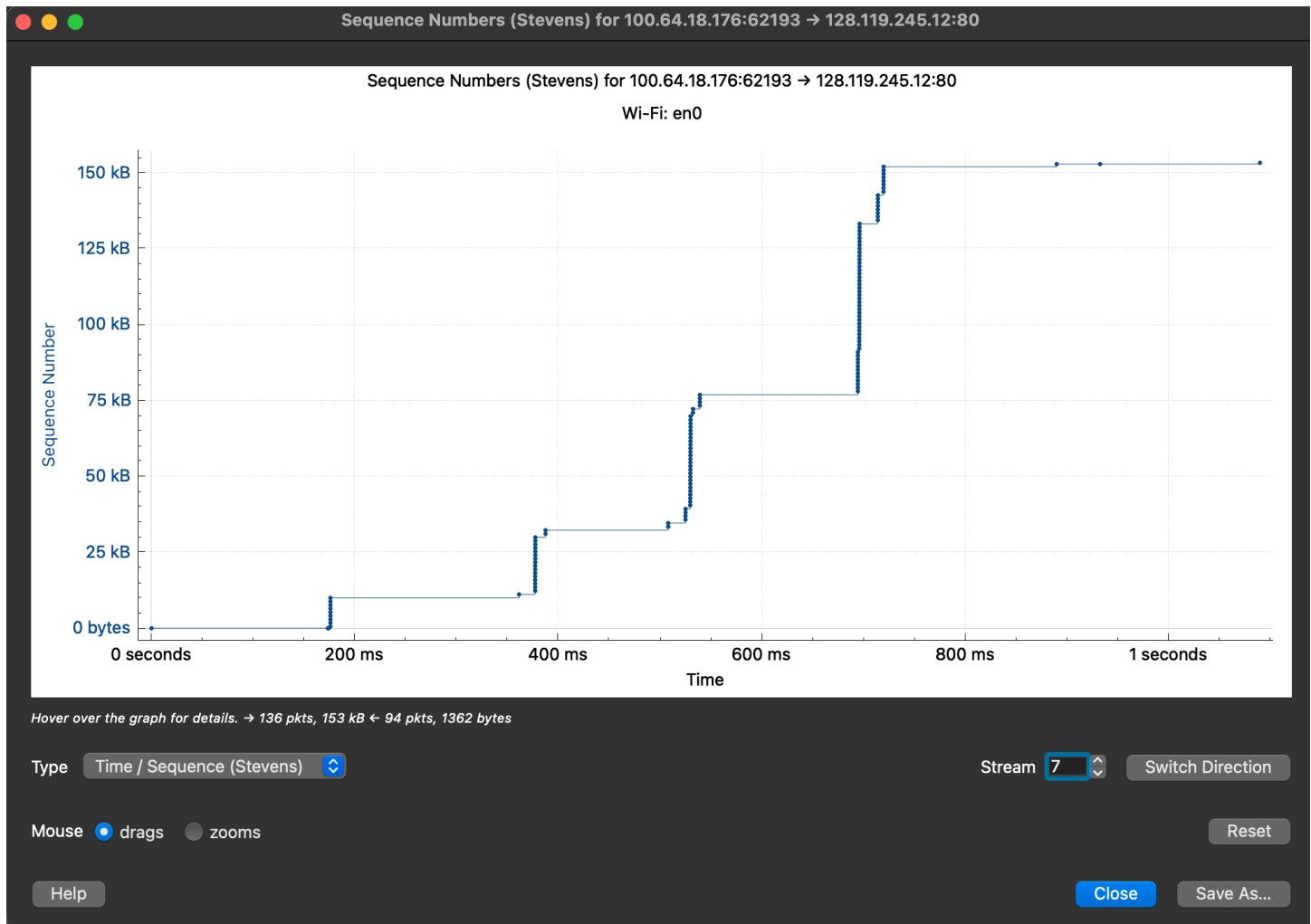
Q11:

Throughput = Total bytes transferred / Total time taken

358	8.163893	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=143820 Win=214016 Len=0 TSval=3000657767 TSecr=3017443365
359	8.163894	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=146168 Win=214016 Len=0 TSval=3000657767 TSecr=3017443371
360	8.163896	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=148516 Win=214016 Len=0 TSval=3000657767 TSecr=3017443371
361	8.163897	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=150864 Win=214016 Len=0 TSval=3000657767 TSecr=3017443371
362	8.163899	128.119.245.12	100.64.18.176	TCP	66	80 → 62193 [ACK] Seq=1 Ack=152913 Win=214016 Len=0 TSval=3000657767 TSecr=3017443371
363	8.163900	128.119.245.12	100.64.18.176	HTTP	846	HTTP/1.1 200 OK (text/html)
364	8.164144	100.64.18.176	128.119.245.12	TCP	66	62193 → 80 [ACK] Seq=152913 Ack=781 Win=130752 Len=0 TSval=3017443541 TSecr=3000657768
365	8.182609	150.171.109.83	100.64.18.176	TCP	66	[TCP ACKed unseen segment] 443 → 62192 [ACK] Seq=1 Ack=2 Win=145 Len=0 TSval=1878011188 TSecr=1391450118
366	8.206801	100.64.18.176	128.119.245.12	HTTP	493	GET /favicon.ico HTTP/1.1
367	8.363810	128.119.245.12	100.64.18.176	HTTP	648	HTTP/1.1 301 Moved Permanently (text/html)
368	8.363989	100.64.18.176	128.119.245.12	TCP	66	62193 → 80 [ACK] Seq=153340 Ack=1363 Win=130496 Len=0 TSval=3017443741 TSecr=3000657965
373	8.401903	100.64.18.176	128.119.245.12	TCP	78	62194 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3900777793 TSecr=0 SACK_PERM
374	8.657270	100.64.18.176	128.119.245.12	TCP	78	62195 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=234061851 TSecr=0 SACK_PERM
375	8.666909	128.119.245.12	100.64.18.176	TCP	74	443 → 62194 [SYN, ACK, ECE] Seq=0 Ack=1 Win=65160 Len=0 MSS=1186 SACK_PERM TSval=3000658246 TSecr=39007779
376	8.667158	100.64.18.176	128.119.245.12	TCP	66	62194 → 443 [ACK] Seq=1 Ack=1 Win=131520 Len=0 TSval=390078058 TSecr=3000658246
377	8.667487	100.64.18.176	128.119.245.12	TCP	1240	62194 → 443 [ACK] Seq=1 Ack=1 Win=131520 Len=1174 TSval=390078058 TSecr=3000658246 [TCP PDU reassembled in
378	8.667504	100.64.18.176	128.119.245.12	TLSv1...	683	Client Hello (SNI=gaia.cs.umass.edu)
379	8.881798	128.119.245.12	100.64.18.176	TCP	74	443 → 62195 [SYN, ACK, ECE] Seq=0 Ack=1 Win=65160 Len=0 MSS=1186 SACK_PERM TSval=3000658427 TSecr=23406185
380	8.882087	100.64.18.176	128.119.245.12	TCP	66	62195 → 443 [ACK] Seq=1 Ack=1 Win=131520 Len=0 TSval=234062076 TSecr=3000658427
381	8.882424	100.64.18.176	128.119.245.12	TCP	1240	62195 → 443 [ACK] Seq=1 Ack=1 Win=131520 Len=1174 TSval=234062076 TSecr=3000658427 [TCP PDU reassembled in
382	8.882438	100.64.18.176	128.119.245.12	TLSv1...	651	Client Hello (SNI=gaia.cs.umass.edu)

Frame 362 8.163899 - 7.448726

Throughput = 213,811 bytes/sec

Q12:

Looking at the Stevens graph, the dots show how the sequence numbers increase over time. I can see that the vertical groups of dots are getting taller every time they appear. This means the sender is sending way more data in each burst than it did before. This kind of fast, exponential growth is a clear sign that TCP is in the Slow Start phase. It's basically the protocol testing the network to see how much data it can handle before it hits a limit.

Q13:

There is a clear pattern where the sender sends a bunch of data and then stops for a bit, creating those horizontal flat lines in the graph. These gaps represent the Round Trip Time (RTT). The sender has to wait for the ACKs to come back from the server before it's allowed to send the next batch of packets. In my graph, this gap is roughly 150ms to 200ms, which is the time it takes for my data to reach the server and for the "all clear" signal to get back to my computer.

Q14:

In my trace, the connection stayed in Slow Start for the entire transfer. You can see this because each "step" in the graph gets taller than the one before it, showing that the sender is still doubling the number of packets it sends. Since the file was only 152KB, it finished before the network got crowded enough to switch to Congestion Avoidance. There were no horizontal flat spots or drops, so no packets were lost.